Research CLI — Agent System Architecture

CLI, core session and routing, tools and research modules, model providers, and embedded runtimes

Dashed frame = architecture zone · Solid frame = subsystem · Box = module and responsibility

PROJECT OVERVIEW

Research CLI is a terminal-first AI agent for code, tools, MCP services, and academic research workflows.
The architecture separates the CLI, core session and routing, tools and research modules, model providers, and embedded runtimes.

CLI FRONTEND · packages/cli

- **research.tsx** — interactive TUI
- **commands/** — mcp.ts · skills.tsx · hooks.tsx
- **ui/commands/research** — arxivCommand · paper · investigate · submit
- **acpSession.ts** — ACP session · stdio transport

CORE ORCHESTRATION · packages/core

Session

- **researchChat.ts** — researchRequest.ts
- **research-config-manager.ts** — academic settings
- **config / memory** — context window

Routing

- **modelRouterService** — routing/modelRouterService.ts
- **defaultStrategy** — routingStrategy.ts
- **classifierStrategy** — gemma · numerical
- **compositeStrategy** — approval · fallback · override

TOOLS & RESEARCH MODULES

Workspace

- **edit.ts · grep.ts** — filesystem changes
- **shell / ls** — local execution

MCP

- **mcp-client.ts** — server sessions
- **mcp-client-manager.ts** — resource routing

analysis/

- **bibliography-manager** — citations
- **experiment-code-generator** — Python · R · Julia
- **research-data-analyzer** — statistics · ML